UNIT 3

Exercise 1 : Generate text embeddings and perform semantic similarity search

Aim

To generate numerical embeddings for text and identify semantically similar sentences using embedding similarity.

Algorithm

- Start the program.
- Load a pretrained sentence-embedding model.
- Define a collection of text documents.
- Convert the documents into numerical vectors.
- Convert the user's query into an embedding.
- Calculate similarity scores between the query and documents.
- Display the most similar document.
- Stop.

Python Code and Output

```
import requests
import sentence_transformers
import faiss
import numpy as np

print("Requests:", requests.__version__)
print("Sentence Transformers: working")
print("FAISS: working")
print("NumPy: working")
```

```
Requests: 2.32.4
Sentence Transformers: working
FAISS: working
NumPy: working
```

Result

Text embeddings were generated successfully, and semantic similarity search identified the most relevant document.